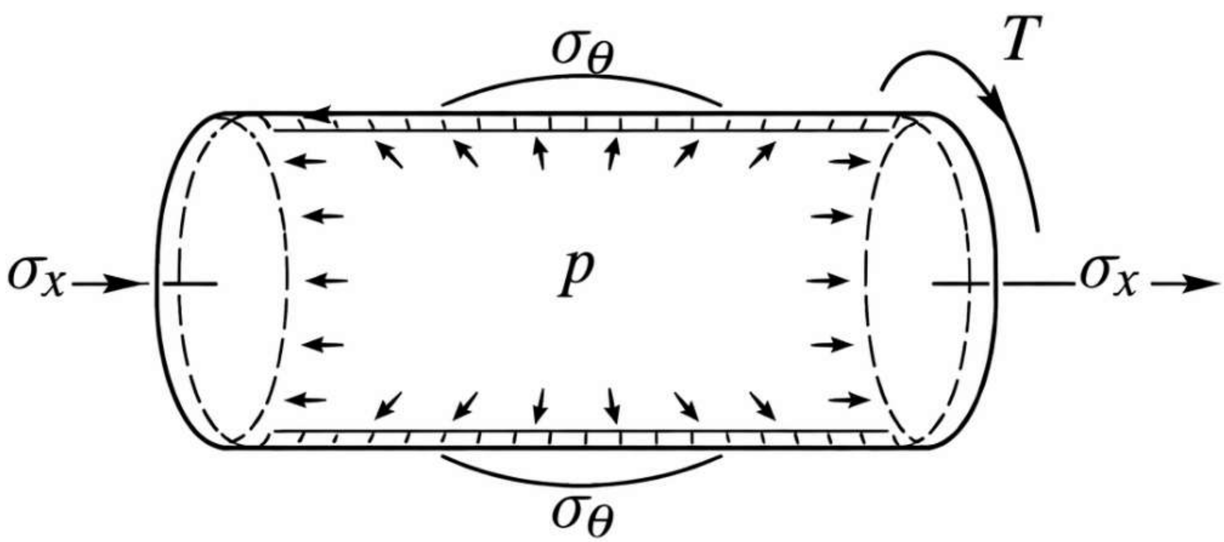
Design and Optimization of a Composite Pressure Vessel Using Classical Laminate Theory and Tsai-Wu Failure Criterion



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Nomenclature

Symbol	Description	Units
P	Internal pressure	[Pa]
T	Applied torque	[N·m]
D	Vessel diameter	[m]
H	Laminate thickness	[m]
T	Single lamina thickness	[m]
n_d	Design factor (safety factor requirement)	[–]
ρ_{comp}	Density of composite material	[kg/m³]
ρ_{Al}	Density of aluminum	[kg/m³]
E_1	Longitudinal modulus of lamina	[Pa]
E_2	Transverse modulus of lamina	[Pa]
G_{12}	In-plane shear modulus	[Pa]
ν_{12}	Major Poisson's ratio	[–]
ν_{21}	Minor Poisson's ratio	[–]
$[Q]$	Reduced stiffness matrix	[Pa]
$[\bar{Q}]$	Transformed reduced stiffness matrix	[Pa]
$[A]$	Extensional stiffness matrix	[N/m]
$[a]$	Compliance matrix	[m/N]
N_x	Axial force resultant per unit length	[N/m]
N_y	Hoop force resultant per unit length	[N/m]
N_s	In-plane shear resultant per unit length	[N/m]
σ_x	Axial stress	[Pa]
σ_y	Hoop stress	[Pa]
τ_{xy}	In-plane shear stress	[Pa]

σ_1, σ_2	Principal stresses	[Pa]
τ_{12}	Principal shear stress	[Pa]
ε_x^0	Midplane axial strain	[–]
ε_y^0	Midplane hoop strain	[–]
γ_{xy}^0	Midplane shear strain	[–]
ε_1, ε_2	Principal strains	[–]
γ_{12}	Principal shear strain	[–]
θ	Ply orientation angle	[degrees]
m, n	Cosine and sine of ply angle (cos θ, sin θ)	[–]
F_1, F_2	Tsai-Wu linear coefficients	[Pa^{-1}]
F_{11}, F_{22}, F_{66}, F_{12}	Tsai-Wu quadratic coefficients	[Pa^{-2}]
S_{fa}	Tsai-Wu safety factor (actual loading)	[–]
S_{fr}	Tsai-Wu safety factor (reversed loading)	[–]
σ_{VM}	von Mises stress	[Pa]
F_{1t}, F_{1c}	Longitudinal tensile/compressive strength	[Pa]
F_{2t}, F_{2c}	Transverse tensile/compressive strength	[Pa]
F_6	Shear strength	[Pa]
A	Cross-sectional area	[m^2]

1 Introduction and approach

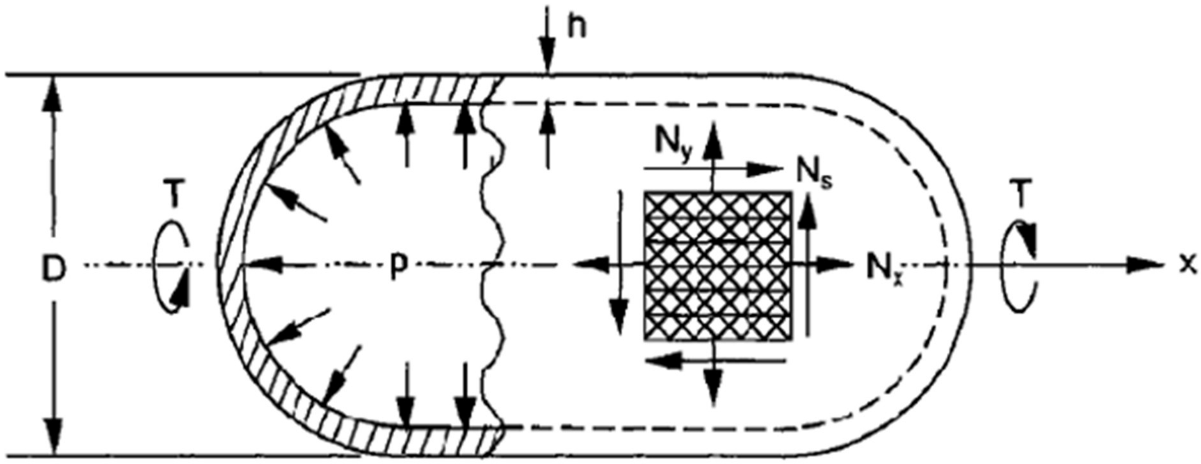

Figure 1.1. Thin wall cylindrical pressure vessel under internal pressure and torque loading

This project focuses on designing a thin-walled cylindrical pressure vessel subjected to loading through internal pressure and applied torque. The goal of which is to develop an optimal laminate layup to satisfy the design specification safety factor while maintaining minimal thickness. The design challenge involves selecting an ideal laminate configuration $[90/\pm45_m]_{ns}$. Where m is the number of ±45 layer groups, and n is the number of repeating sequences there are. The vessel must withstand 2.07MPa of internal pressure and 283kN*m of torque. The outer diameter is 89cm with a safety factor of 2. A comparison will be made between the reference aluminum and the composite layup as to benefits in weight reduction while maintaining structural integrity. The project will cover composite laminate theory (CLT), stress transformations, and failure criteria to configure the most efficient solution validated by a comparison against baseline aluminum, ultimately highlighting the advantages of fiber reinforced composites in multi-load pressure vessel applications.

2 Stress Analysis

Before any important calculations can be made, the state of stress of the composite pressure vessel must be determined. The vessel is subjected to internal pressure and torque loading which contribute to the state of stress. Stress analysis can be applied to determine the in-plane stress resultants acting on a laminate element due to the combined loading.

2.1 Applied Loads and Geometry

Table 1 in the Appendix contains the design conditions of the cylindrical, thin-walled pressure vessel depicted in Figure 1. Although it is impossible to calculate a numerical value for the state of stress without the laminate thickness, we can instead calculate the stress state in reference to the unit loads and laminate thickness.

2.2 Unit Load Resultants

The unit load resultants represent the unit loads acting on an element of the cylindrical shell along axial and hoop directions. These values can be used to gain a general understanding of the state of stress without knowing the laminate thickness and can be used in the future when calculating in-plane strain while determining Tsai-Wu safety factor. They will also be useful in determining key information about the aluminum vessel. Equations (2.2.1-2.2.3) detail how the unit loads are calculated as well as how they relate to the state of stress of the vessel. Using the data found in Table 1 in the Appendix, the states of stress are calculated in the MATLAB script found in Appendix 3.

$$\mathrm{N_x} = \bar{\sigma}_{\mathrm{x}} h = \frac{\mathrm{pD}}{4} = 460\ kN/m \tag{2.2.1}$$

$$\mathrm{N_y} = \bar{\sigma}_{\mathrm{y}} h = \frac{\mathrm{pD}}{2} = 920\ kN/m \tag{2.2.2}$$

$$\mathrm{N_s} = \bar{\tau}_{\mathrm{s}} h = \frac{2\mathrm{D}}{\pi \mathrm{D}^2} = 228 kN/m \tag{2.2.3}$$

2.3) State of Stress

Although unnecessary for Tsai-Wu criterion, equations (2.2.1-2.2.3) can be rearranged to solve for the state of stress of a vessel under the given combined loading. Table 2.3.1 contains the state of stress for this vessel. These expressions will be utilized in the future when calculating the thickness of the aluminum vessel.

Table 2.3.1. State of stress of representative element for given loading.

Axial Stress	Hoop Stress	In-plane Shear Stress
$\bar{\sigma}_{\mathrm{x}} = \frac{460}{h} kN/m^2$	$\bar{\sigma}_{\mathrm{y}} = \frac{920}{h} kN/m^2$	$\bar{\tau}_{\mathrm{s}} = \frac{228}{h} kN/m^2$

3 Evaluation of Laminate Properties and Safety Factor

To determine the optimal layup for the composite pressure vessel, the MATLAB script and function found in Appendices 3 and 4 were used to generate and evaluate layups in the design space of $[90/\pm 45_m]_{ns}$. Following the flowchart from section 9.5 of the textbook found in Figure 3.1, this process identifies the thinnest laminate while maintaining a Tsai-Wu safety factor above 2.

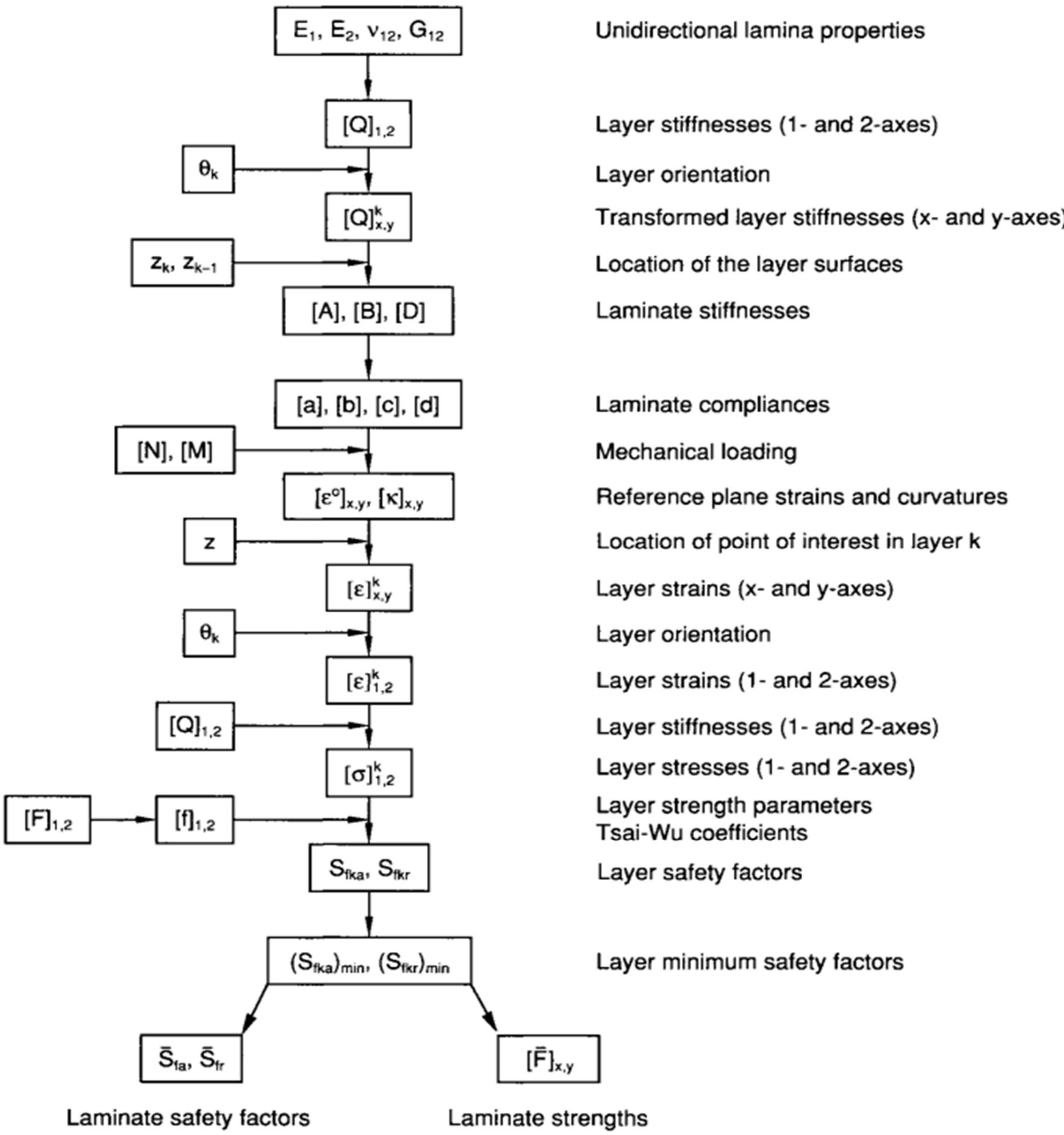

Figure 3.1: Flowchart for stress and failure analysis of multidirectional laminates

3.1) Laminate Layup Generation

Before evaluating each layup, the stiffness matrix [Q] is calculated based on known lamina properties using equations (3.1.1-3.1.5).

$$Q_{11} = \frac{E_1}{1 - \nu_{12}\nu_{21}} \tag{3.1.1}$$

$$Q_{22} = \frac{E_2}{1 - \nu_{12}\nu_{21}} \tag{3.1.2}$$

$$Q_{12} = Q_{21} = \frac{\nu_{21}E_1}{1 - \nu_{12}\nu_{21}} \tag{3.1.3}$$

$$Q_{66} = G_{12} \tag{3.1.4}$$

$$Q = \begin{bmatrix} Q_{11} & Q_{12} & 0 \\ Q_{21} & Q_{22} & 0 \\ 0 & 0 & Q_{66} \end{bmatrix} \tag{3.1.5}$$

Using a nested for loop, the script iterates for each value of m and n from 1 to 10. This ensures that each layup in the design space is evaluated. Equations (3.1.6) and (3.1.7) are used to calculate the number of plies and then the total thickness of the layup, respectively, where the variable t represents the thickness of each lamina layer.

$$NumPlies = 2n(2m + 1) \tag{3.1.6}$$

$$h = NumPlies * t \tag{3.1.7}$$

Once these values have been calculated, the script generates the layup using MATLAB's "repmat" and "fliplr" functions. These built in functions allow us to create and evaluate the layup based on the value of m and n we need to evaluate.

3.2) Stiffness Matrix and Midplane strains

Referencing the flowchart in Figure 3.1, the next steps require us to find the transformed layer stiffnesses, location of the later surfaces, and laminate stiffnesses to find the [a] matrix used to calculate the laminate stiffnesses and in-plane strains. All of this is done using the MATLAB function "laminateStiffness" that we created in Appendix 4. This function takes the known lamina stiffness properties E1, E2, G12, and v12 as well as the lamina layer thickness and the layup generated previously to calculate the [a] matrix. The function starts by calculating the transformed reduced stiffness matrix using equations (3.2.1-3.2.6). Using these values, it uses equations (3.2.7) to calculate the [A] matrix. Because the laminate is symmetric, the [a] matrix can be calculated by simply taking the inverse of the [A] matrix.

$$Q_{xx} = m^4Q_{11} + n^4Q_{22} + 2m^2n^2Q_{12} + 4m^2n^2Q_{66} \tag{3.2.1}$$

$$Q_{yy} = n^4Q_{11} + m^4Q_{22} + 2m^2n^2Q_{12} + 4m^2n^2Q_{66} \tag{3.2.2}$$

$$Q_{xy} = m^2n^2Q_{11} + m^2n^2Q_{22} + (m^4 + n^4)Q_{12} - 4m^2n^2Q_{66} \tag{3.2.3}$$

$$Q_{xs} = m^3nQ_{11} - mn^3Q_{22} - mn(m^2 - n^2)Q_{12} - 2mn(m^2 - n^2)Q_{66} \tag{3.2.4}$$

$$Q_{ys} = mn^3Q_{11} - m^3nQ_{22} + mn(m^2 - n^2)Q_{12} + 2mn(m^2 - n^2)Q_{66} \tag{3.2.5}$$

$$Q_{ss} = m^2n^2Q_{11} + m^2n^2Q_{22} - 2m^2n^2Q_{12} + (m^2 - n^2)^2Q_{66} \tag{3.2.6}$$

$$A_{ij} = \sum_{k=1}^{n} Q_{ij}^{k}(z_k - z_{k-1}) \tag{3.2.7}$$

After the [a] matrix is calculated, the effective laminate moduli are calculated using equations (3.2.8-3.2.10), which can be applied in cases of symmetric laminates.

$$\overline{\mathrm{E}}_{\mathrm{x}} = \frac{1}{\mathrm{h*a(1,1)}} \tag{3.2.8}$$

$$\overline{\mathrm{E}}_{\mathrm{y}} = \frac{1}{\mathrm{h*a(2,2)}} \tag{3.2.9}$$

$$\overline{G}_{xy} = \frac{1}{h^*a(3,3)} \tag{3.2.10}$$

This gives us the effective laminate properties of the specific layup being tested. The longitudinal elastic modulus is represented by $\overline{E}_x$, transverse elastic modulus by $\overline{E}_y$, and the in-plane shear modulus by $\overline{G}_{xy}$.

Equation (3.2.11) calculates the in-plane strains by multiplying the [a] matrix by the 3x1 N matrix which contains the axial (N_x), hoop (N_y), and shear (N_s) unit loads experienced by the vessel due to the applied pressure and torque, which were calculated in section 2. This yields a 3x1 matrix that contains the in-plane strains ε_x^0, ε_y^0, and γ_s^0. This equation is applicable due to the symmetry of the layup.

$$\varepsilon^0 = [a] * [N] \tag{3.2.11}$$

3.3) Ply-Level Stress and Strain Calculation

Once the in-plane strains have been calculated, a for loop that runs for each ply in the layup is created to evaluate the stress and strain of each ply before calculating the factor of safety of each ply. Equation (3.3.1) is then used to transform the in-plane strains to principal strains using the strain transformation matrix [T], defined in equation (3.3.2) where c is equal to cos(θ), and s is equal to sin(θ). Equation (3.3.1) yields a 3x1 matrix which contains the principal strains ε_1, ε_1, and γ_6.

$$\varepsilon_{12} = [T] * [\varepsilon^0] \tag{3.3.1}$$

$$T = \begin{bmatrix} c^2 & s^2 & cs \\ s^2 & c^2 & -cs \\ -2cs & 2cs & c^2 - s^2 \end{bmatrix} \tag{3.3.2}$$

Using the principal strains (ε_{12}), the principal stresses can be calculated using equation (3.3.3), which multiplies the principal strain matrix by the stiffness matrix calculated in section 3.1 using equation (3.1.5). This yields a 3x1 matrix containing the principal stresses σ_1, σ_2, and τ_{12}.

$$\sigma_{12} = [Q] * \varepsilon_{12} \tag{3.3.3}$$

3.4) Tsai-Wu Failure Criterion and Safety Factor

Under the same for loop as the calculations in section 3.3, the Tsai-Wu safety factor of each ply can be calculated. The Tsai-Wu factor of safety is calculated using equations (3.4.1) and (3.4.2), where a and b are defined by equations (3.4.3) and equation (3.4.4), respectively. The variable S_{fa} represents the factor of safety for the actual state of stress and S_{fr} represents the factor of safety under the same stress with a reversed sign.

$$S_{fa} = \frac{-b + \sqrt{b^2 + 4a}}{2a} \tag{3.4.1}$$

$$S_{fr} = \left| \frac{-b - \sqrt{b^2 + 4a}}{2a} \right| \tag{3.4.2}$$

$$a = F_{11}\sigma_1^2 + F_{22}\sigma_2^2 + F_{66}\tau_{12}^2 + 2F_{12}\sigma_1\sigma_2 \tag{3.4.3}$$

$$b = F_1\sigma_1 + F_2\sigma_2 \tag{3.4.4}$$

Before the a and b variables can be calculated, the Tsai-Wu coefficients found in equations (3.4.1-3.4.4) must be determined. Using equations (3.4.5-3.4.10), the Tsai-Wu coefficients can be calculated based on the known failure strengths of the composite found in Table 2 in the appendix.

$$F_1 = \frac{1}{F_{1t}} - \frac{1}{F_{1c}} \tag{3.4.5}$$

$$F_2 = \frac{1}{F_{2t}} - \frac{1}{F_{2c}} \tag{3.4.6}$$

$$F_{11} = \frac{1}{F_{1t}F_{1c}} \tag{3.4.7}$$

$$F_{22} = \frac{1}{F_{2t}F_{2c}} \tag{3.4.8}$$

$$F_{66} = \frac{1}{F_6^2} \tag{3.4.9}$$

$$F_{12} = -\frac{1}{2}(F_{11}F_{22})^{1/2} \tag{3.4.10}$$

The for loop terminates after calculating the factor of safety. The minimum ply-level factor of safety is recorded as the laminate's overall factor of safety.

3.5) Optimization Criterion

At this point in the script, the number of plies, total thickness, effective laminate moduli, and Tsai-Wu safety factor of the laminate have all been calculated for each combination of values for m and n. An if statement is used to record these values if safety factor is greater than 2, and if the total thickness is less than the previous minimum thickness. Once the nested for loop mentioned section 3.1 has run for all combinations of m and n, the final values stored will be the optimal layup that minimizes total thickness while maintaining a safety factor above 2.

3.6) Aluminum Vessel Comparison

Once the optimal layup for the composite vessel has been found, it is useful to compare it to a reference aluminum vessel of the same strength to determine how much weight would be saved using the composite vessel. First the thickness of the aluminum vessel must be determined using the von Mises Criterion using the same design factor of 2. The aluminum has a material yield strength of $\sigma_y = 242MPa$ and a density of $\rho_{Al} = 2.8g/cm^3$. The von Mises yield criterion is given in equation (3.6.1).

$$\sigma_{VM} = \frac{\sqrt{2}\sigma_y}{n_d} \tag{3.6.1}$$

Because the pressure is thin-walled, radial stress is negligible and the von Mises stress can be calculated using equation (3.6.2). The variables $\bar{\sigma}_x$, $\bar{\sigma}_y$, and $\bar{\tau}_s$ represent components of the state of stress of the vessel derived in section 2.3 and found in Table 2.3.1.

$$\sigma_{VM} = \sqrt{\bar{\sigma}_x^2 - \bar{\sigma}_x\bar{\sigma}_y + 3\bar{\tau}_s^2} \tag{3.6.2}$$

By substituting the known values for yield strength and design factor as well as the values for the state of stress found in Table 2.3.1, the MATLAB code can calculate the thickness of the aluminum vessel.

With the thickness of the aluminum and composite vessel calculated, the weight to length ratio of each vessel can be determined by using equation (3.6.3) to calculate the cross-sectional area of each vessel, then multiplying each area by its respective density. Using these calculations the MATLAB code then calculates the compared weight by dividing the weight ratio of the composite by the weight ratio of the aluminum.

$$A = \left[\left(\frac{D}{2}\right)^2 - \left(\frac{D}{2} - h\right)^2\right]\pi \tag{3.6.3}$$

3.7) Summary

Table 3.7.1 contains the optimal layup structure as well as important data and calculations associated with the layup. The effective laminate moduli are also included as well as the composite vessel's weight compared to the aluminum vessel.

Table 3.7.1. Optimal layup structure details.

Optimal Layup Structure	$[90/\pm45_1]_{6s}$
Laminate Thickness (h) [mm]	4.572
Tsai-Wu Safety Factor	2.01
Longitudinal Elastic Modulus $(\overline{\mathrm{E}}_{\mathrm{x}})$ [GPa]	28.57
Transverse Elastic Modulus $(\overline{\mathrm{E}}_{\mathrm{y}})$ [GPa]	65.54
In-plane Shear Modulus $(\overline{\mathrm{G}}_{\mathrm{xy}})$ [GPa]	27.75
Number of Plies	36
Weight Savings compared to Aluminum Vessel	69.23%

4 Discussion and Conclusion

As seen in Table 3.7.1, the optimal values of m and n were found to be 1 and 6, respectively. This means that the optimal composite layup was identified to be $[90/\pm45_1]_{6s}$. This layup configuration consists of 36 total plies and a total laminate thickness of 4.572 mm. It maintains a Tsai-Wu safety factor of 2.01, which meets the design requirement of a design factor of at least 2, meaning the designed pressure vessel could withstand the combined loading dictated by Figure 1.1.

The effective moduli calculated are given in Table 3.7.1. At first glance, these numbers seem to be incorrect. One typically expects the longitudinal modulus to be much larger than the transverse and shear moduli. However, the composite layup given in Table 3.7.1 can be referenced to understand these values. The layup contains no 0-degree plies and many 45 and 90-degree plies. This information supports the resulting effective moduli as 0-degree plies are typically responsible for contributing to longitudinal stiffness. The surplus in 90-degree plies explains the larger transverse modulus and the surplus in 45-degree plies explains the larger in-plane shear modulus.

Compared to the reference aluminum vessel, the composite design exhibits a 69.23% reduction in weight while also achieving a 37.2% reduction in wall thickness. Aluminum is traditionally favored for pressure vessels due to ease of fabrication, but the tailored anisotropy of the composite laminate allows it to meet strength requirements more efficiently by aligning fiber directions with principal stress directions.

This portrays the advantage of composite materials in structural design. Directional strength can be engineered into the layup. In this case, the $[90/\pm45_1]_{6s}$ layup leveraged this by distributing fibers to handle the multiaxial stress state effectively.

The design process for the composite pressure vessel combined CLT with pressure vessel mechanics to draw values of the state of stress $(\sigma_x, \sigma_y, \tau_{xy})$ and calculated the effective laminate moduli (E_x, E_y, G_{xy}). Then using Tsai-Wu failure criterion accounting for multi-axial stress we can optimize for the ideal $[90/\pm45_m]_{ns}$ configuration that satisfies the design factor. Using iterative optimization of the layup template the program designed a layup which has the smallest and lesser mass profile while maintaining the factor of safety.

5 Appendix

Table 1: Design Specification Table

Property/Constraint	Design Specification
Internal pressure, p	2.07 MPa (300 psi)
Torque loading, T	283 kN-m (2.5×10^6 lb-in)
Diameter, D	89 cm (35 in)
Design Factor, n_d	2

Table 2: Mechanical Properties of Carbon/Epoxy Composite Lamina

Lamina Property	Value
Longitudinal modulus, E_1, GPa (Msi)	147 (21.3)
Transverse modulus, E_2, GPa (Msi)	10.3 (1.50)
Shear modulus, G_{12}, GPa (Msi)	7.0 (1.00)
Major Poisson's ratio, ν_{12}	0.27
Major Poisson's ratio, ν_{21}	0.02
Longitudinal tensile strength, F_{1t}, MPa (ksi)	2280 (330)
Transverse tensile strength, F_{2t}, MPa (ksi)	57 (8.3)
In-plane shear strength, F_6, MPa (ksi)	76 (11.0)
Longitudinal compressive strength, F_{1c}, MPa (ksi)	1725 (250)
Transverse compressive strength, F_{2c}, MPa (ksi)	228 (33)
Lamina Layer Thickness, mm (in)	0.127 (0.005)
Density, ρ, g/cm^3 (lb/in^3)	1.38 (0.050)

Appendix 3: MATLAB script used for all major calculations

```
% Design specifications

p=2.07*10^6; % internal pressure [Pa]

T=283*10^3; % Torque Loading [N-m]

D=0.89; % Diameter [m]

nd=2; % design factor

rho_comp=1380; %density [kg/m^3]

% basic lamina properties:

E1=147*10^9; % Pa

E2=10.3*10^9; % Pa

G12=7*10^9; % Pa

v12=0.27;

v21=0.02;

t=0.127*(10^-3); % lamina layer thickness [m]

% Failure Strengths for composite [Pa]:

F1t=2280*10^6;

F1c=1725*10^6;

F2t=57*10^6;

F2c=228*10^6;

F6=76*10^6;

% Tsai-Wu Coefficients:

F1=1/F1t-1/F1c;

F2=1/F2t-1/F2c;

F11=1/(F1t*F1c);

F22=1/(F2t*F2c);

F66=1/(F6^2);

F12=-0.5*sqrt(F11*F22);

% Aluminum Vessel:

Fy=242*10^6; % yield strength [Pa]

rho_Al=2800; % density (kg/m^3)
```

```
% Q matrix calculation
Q11=E1/(1-v12*v21);
Q22=E2/(1-v12*v21);
Q12=v21*E1/(1-v12*v21);
Q21=Q12;
Q66=G12;
% calculates unit loads (equation 9.64)
Nx=(p*D)/4;
Ny=(p*D)/2;
Ns=(2*T)/(pi*D^2);
N=[Nx;Ny;Ns]; % [N/m]
% initialize variables
min_h=Inf;
optimal_m=NaN;
optimal_n=NaN;
Max_m=10;
Max_n=10;
% for loops run for each value of m and n
for m=1:Max_m
  for n=1:Max_n
     Num_Plies=2*n*(2*m+1); % calculate number of plies
     h=Num_Plies*t; % calculate total thickness [m]
     % generates layup based on m and n
     Layup=repmat([90,repmat([45,-45],1,m)],1,n);
     Layup=[Layup,fliplr(Layup)];
     % uses function to calculate [a]
     a=laminateStiffness(E1,E2,G12,v12,t,Layup);

     % eqn 7.84:
     E_x=1/(h*a(1,1)); % uses [a] to find E_x
```

```
E_y=1/(h*a(2,2)); % uses [a] to find E_y
G_xy=1/(h*a(3,3)); % uses [a] to find G_xy
% Module 8, slide 2
Epsilon_0=a*N; % in-plane strains
Sigma=zeros(3,1,Num_Plies);
Sigma12=zeros(3,1,Num_Plies);
Sfa=zeros(1,Num_Plies);
Sfr=zeros(1,Num_Plies);
for i=1:1:Num_Plies % for loop runs for each ply
    % convert angles to rads
    theta=Layup(i)*pi/180;
    c=cos(theta);
    s=sin(theta);

    % Strain Transformation matrix:
    T=[c^2 s^2 c*s;s^2 c^2 -c*s;-2*c*s 2*c*s c^2-s^2];
    % Transforms in-plane strains to principal strains
    Epsilon_12=T*Epsilon_0;
    % Uses Principal strains and Q to calculate principal stresses
    Sigma12(:,i)=[Q11 Q12 0; Q21 Q22 0; 0 0 Q66]*Epsilon_12;
    % Tsa-Wu:
    s1=Sigma12(1,i);
    s2=Sigma12(2,i);
    t12=Sigma12(3,i);
    A_tsaiwu=F11*s1^2+F22*s2^2+F66*t12^2+2*F12*s1*s2;
    B=F1*s1+F2*s2;

    % Tsai wu factor of safety for ply
    Sfa(i)=(-B+sqrt(B^2+4*A_tsaiwu))/(2*A_tsaiwu); % actual loading
    Sfr(i)=abs((-B-sqrt(B^2+4*A_tsaiwu))/(2*A_tsaiwu)); % reversed loading
```

```
        end
        % Tsai Wu safety factor for entire laminate
        Sf=min([Sfa,Sfr]);
          % stores laminate information for laminate with lowest h value if Sf >=0
        if Sf>=2 && h<min_h
          min_h=h;
          optimal_m=m;
          optimal_n=n;
          best_Sf=Sf;
          best_Layup=Layup;
          Optimal_Plies=Num_Plies;
          Optimal_E_x=E_x;
          Optimal_E_y=E_y;
          Optimal_G_xy=G_xy;
          Optimal_a=a;
        end

    end
end
%% Weight comparison:
ha=1257000/((sqrt(2)*Fy)/nd); % uses von mises to solve for aluminum thickness [m]
A_al=((D/2)^2-(D/2-ha)^2)*pi; % cross sectional area of Al vessel [m^2]
A_comp=((D/2)^2-(D/2-min_h)^2)*pi; % cross sectional area of composite vessel [m^2]
W_al=rho_Al*A_al; % weight ratio of aluminum [kg/m]
W_Comp=rho_comp*A_comp; % weight ratio of Composite [kg/m]
Weight_Compare=W_Comp/W_al;
%% Output
fprintf("Optimal Layup: [90/(±45)%d]%ds\n",optimal_m,optimal_n)
```

```
fprintf("Minimum h = %.3f mm with S_f = %.2f\n",min_h*1000,best_Sf)
fprintf('Effective Laminate Properties:\n')
fprintf('Longitudinal elastic modulus: %.2f GPa\n',Optimal_E_x/1e9)
fprintf('Transverse elastic modulus: %.2f GPa\n',Optimal_E_y/1e9)
fprintf('In-plane shear modulus: %.2f GPa\n',Optimal_G_xy/1e9)
fprintf('Number of Plies: %.2f Plies\n',Optimal_Plies)
fprintf("The weight of the composite vessel is equal to %.2f%% of the aluminum vessel.\n",Weight_Compare*100)
```

Appendix 4: Matlab function used to calculate [a] matrix

```
function [a] = laminateStiffness(E1, E2, G12, v12, t, Layup)
  % Check if thickness is a scalar or array and validate
  nLayers = length(Layup);
  if isscalar(t)
     t = t * ones(1, nLayers);
  else
     if length(t) ~= nLayers
        error('Thickness array length must match the number of layers.');
     end
```

```
end

% Total thickness of the laminate
H = sum(t);

% Compute v21 and Q matrix components
v21 = (E2 / E1) * v12;
denom = 1 - v12 * v21;
Q11 = E1 / denom;
Q12 = (v12 * E2) / denom;
Q22 = E2 / denom;
Q66 = G12;

% Initialize A, B, D matrices
A = zeros(3);
B = zeros(3);
D = zeros(3);

% Initialize z_position at the bottom of the laminate
z_position = -H / 2;

for k = 1:nLayers
    theta = Layup(k);
    theta_rad = theta * pi / 180;
    m = cos(theta_rad);
    n = sin(theta_rad);

    % Compute Qbar components
    m2 = m^2;
    n2 = n^2;
```

```
mn = m * n;
m4 = m^4;
n4 = n^4;
m3n = m^3 * n;
mn3 = m * n^3;
m2n2 = m2 * n2;

Qbar11 = Q11*m4 + 2*(Q12 + 2*Q66)*m2n2 + Q22*n4;
Qbar12 = (Q11 + Q22 - 4*Q66)*m2n2 + Q12*(m4 + n4);
Qbar22 = Q11*n4 + 2*(Q12 + 2*Q66)*m2n2 + Q22*m4;
Qbar16 = (Q11 - Q12 - 2*Q66)*m3n + (Q12 - Q22 + 2*Q66)*mn3;
Qbar26 = (Q11 - Q12 - 2*Q66)*mn3 + (Q12 - Q22 + 2*Q66)*m3n;
Qbar66 = (Q11 + Q22 - 2*Q12 - 2*Q66)*m2n2 + Q66*(m4 + n4);

Qbar = [Qbar11, Qbar12, Qbar16;
    Qbar12, Qbar22, Qbar26;
    Qbar16, Qbar26, Qbar66];

% Layer thickness
tk = t(k);

% Midplane position of current layer
z_mid = z_position + tk / 2;
z_k = z_mid;  % Midplane relative to laminate midplane

% Update A, B, D matrices
A = A + Qbar * tk;
B = B + Qbar * tk * z_k;
D = D + Qbar * (tk^3 / 12 + tk * z_k^2);
```

```
        % Move to the bottom of the next layer
        z_position = z_position + tk;
    end

    a=inv(A);
end
```